RetainFlow Automation: Customer Churn Prediction

Data Science Bootcamp | Capstone Project | Sohee Lee

Dataset Overview

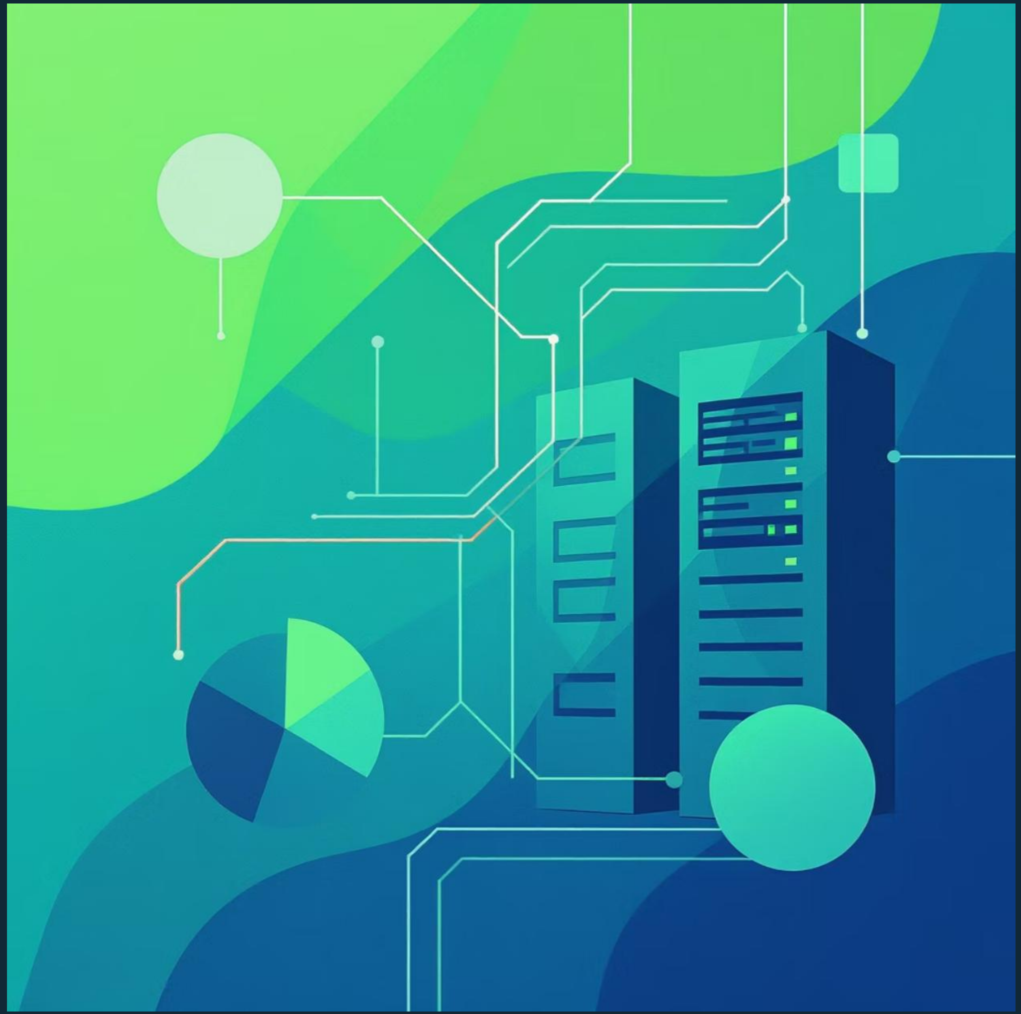
7,000+ Customers

Telco industry dataset

Multi-dimensional Features

- Contract terms & payment methods
- Service usage patterns
- Billing history & charges and so on

⚠ **Challenge:** Class imbalance & limited behavioral signals



Why Churn Matters

Business Impact

Fierce competition → churn = direct revenue loss

New customer = 5x more expensive than retention (HBR)

5% churn reduction = +25-95% profit(Bain&Co)

Churn Prediction + Automation

Real-time retention triggers when customers show churn signals

Churn Risk Assessment & Automation

1 Churn Model Assessment

Applied churn model to identify at-risk customers

2 Revenue Integration

Combined with revenue model for comprehensive risk analysis

3 High-Risk Identification

Pinpointed customers with highest churn probability

4 MVP Automation

Developed simple automation for implementation

Key Data Insights

High-Risk Profile

Short tenure + month-to-month contracts + e-check payments

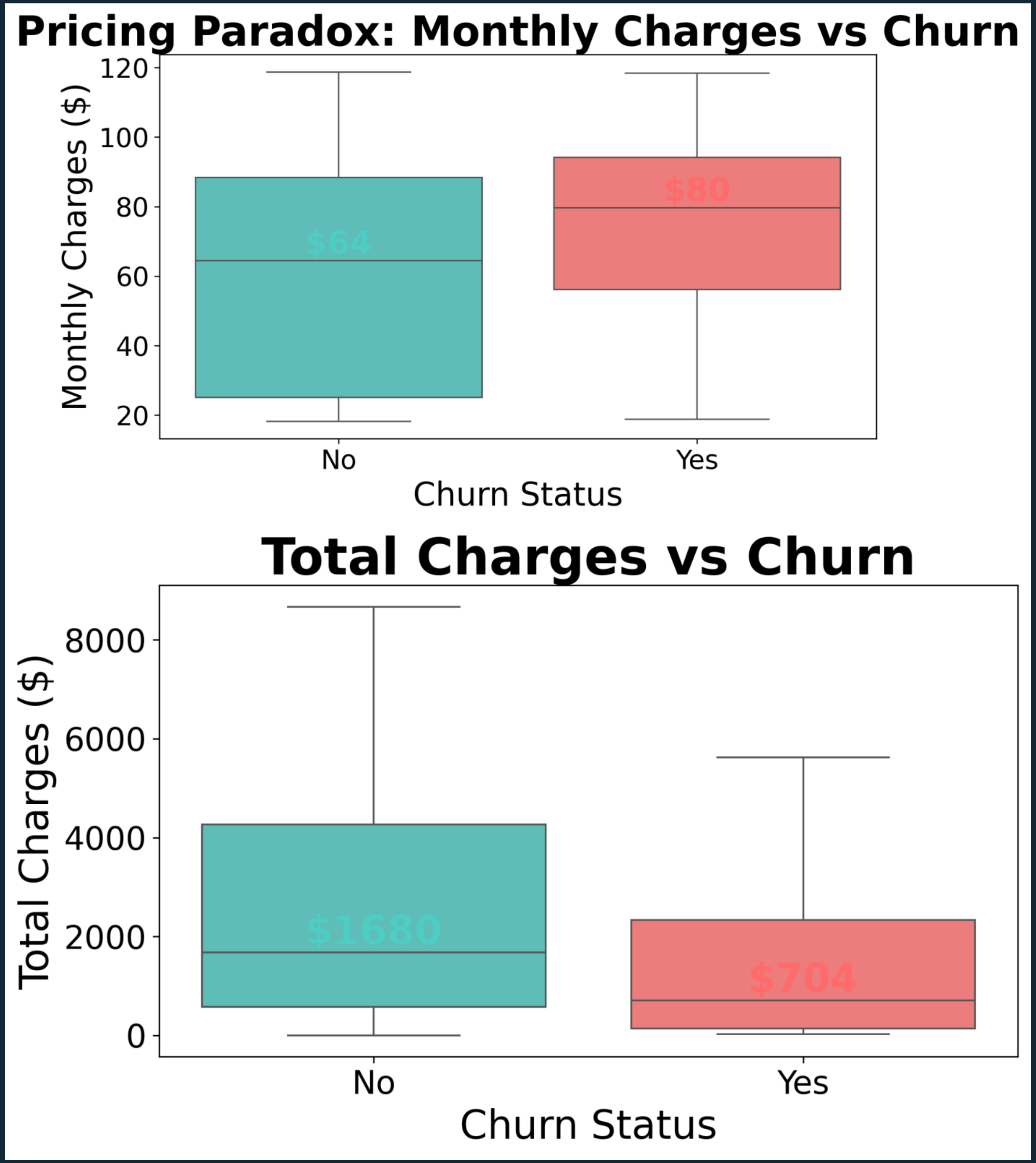
Service Gaps Drive Churn

No tech support or online security = higher churn

Pricing Paradox

High monthly charges ↑ churn

High total charges ↓ churn



Churn Modeling Results

- Key Churn Drivers

Contract Month-to-Month

0.108 importance - Most critical factor

Electronic Check Payment

0.085 importance - Second highest risk

No Online Security

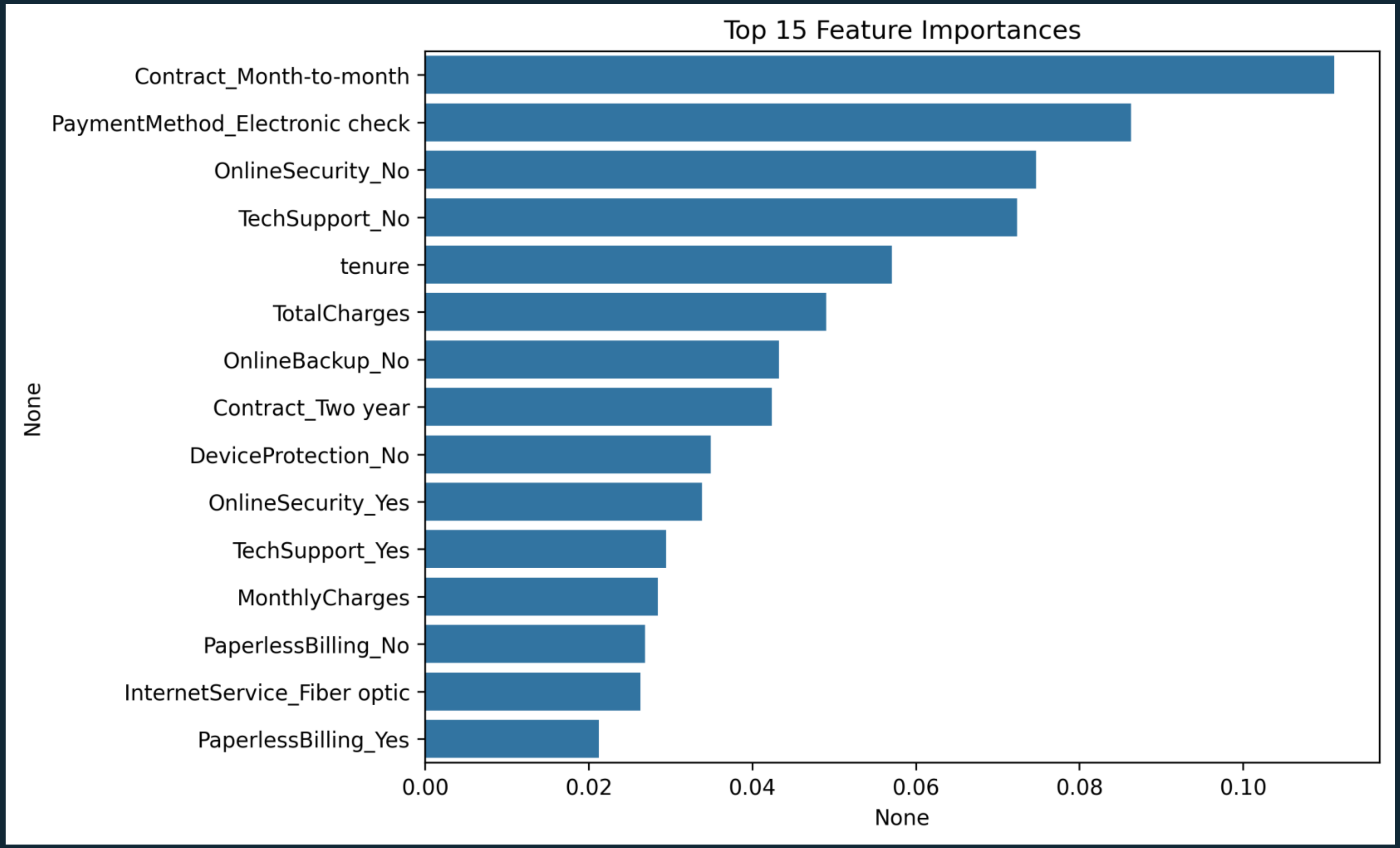
0.072 importance - Third risk factor

No Tech Support

0.068 importance - Fourth risk factor

Short-term Contract + Unstable Payment = Churn Risk

No Security Services = Churn Risk



Churn Modeling Results

- Model Performance Trade-offs

Threshold = 0.5 (Precision = 64%, Recall = 53%)

High Precision

Low Recall

Many churners missed

Threshold = 0.3 (Precision = 55%, Recall = 72%)

Higher Recall

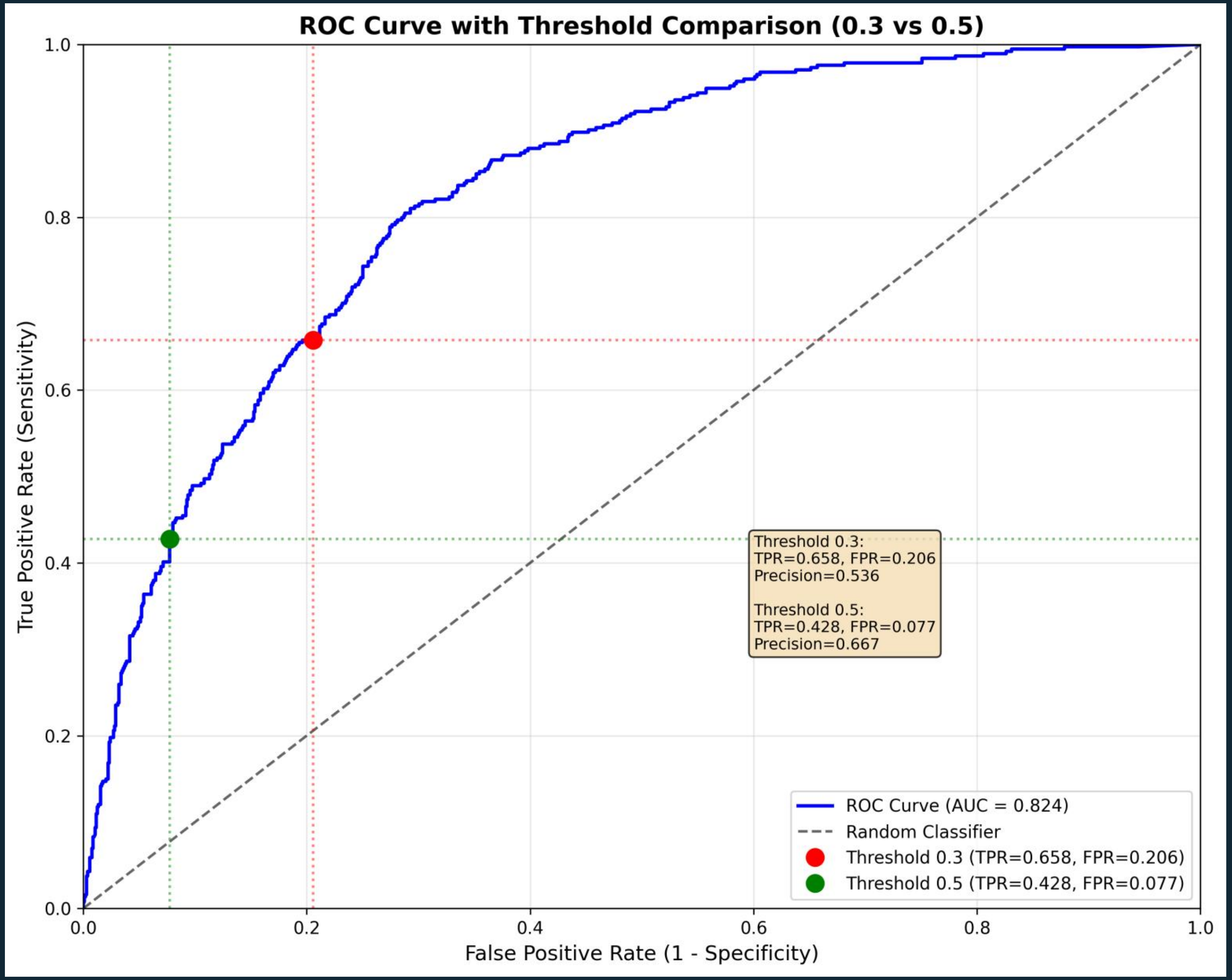
Lower Precision

More false alarms

0.81

ROC AUC Score

Strong model performance



> Trade-off: Recall ↔ Precision → depends on business goals

Churn Modeling Results

- Customer segmentation

High Value + Low Risk

VIP Customers - Premium service delivery focus

Medium Value + Low Risk

Potential Customers - Upselling opportunities

Low Value + Low Risk

Stable Customers - Maintain basic services

High Value + High Risk

At-Risk Customers - Immediate retention needed

Monthly charges analyzed against churn probability to enable tailored retention strategies

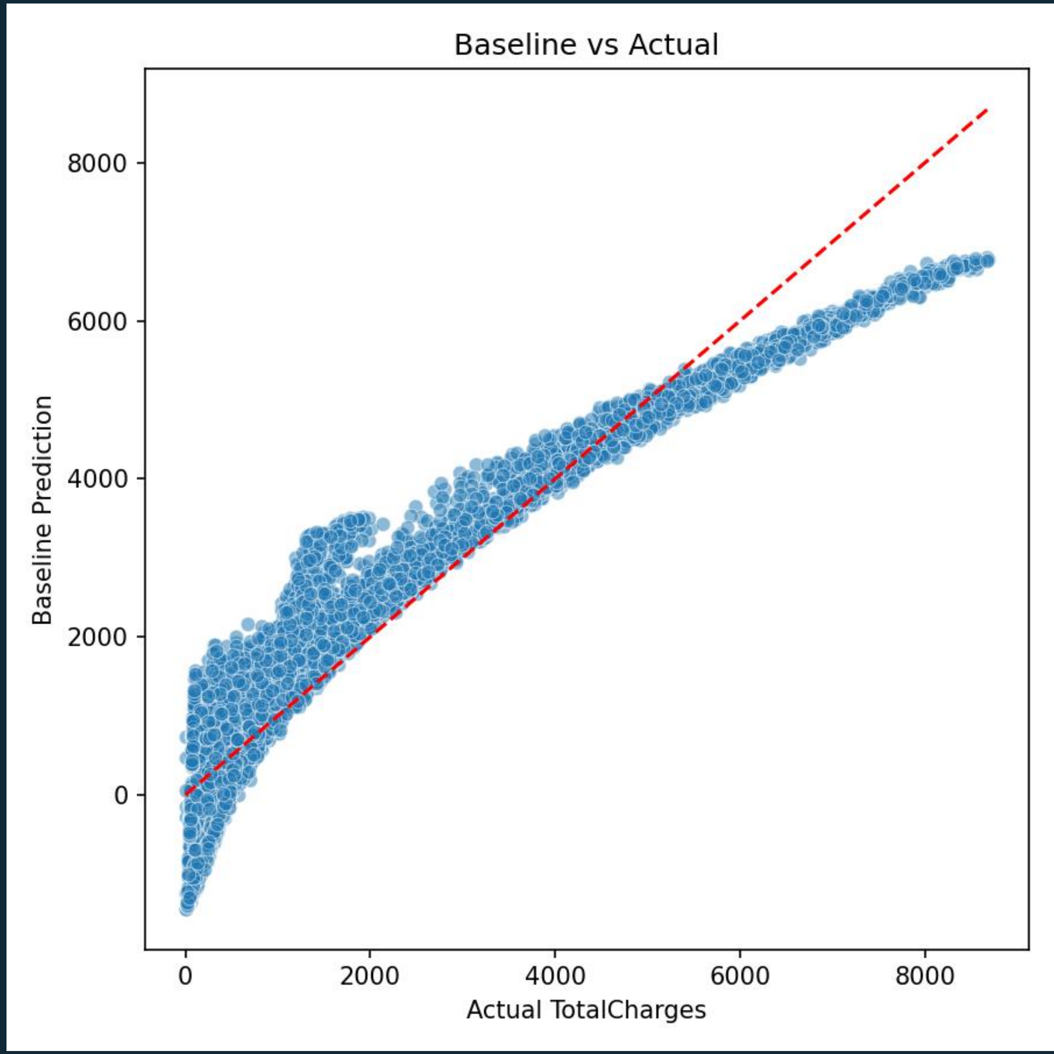
Revenue Modeling Results

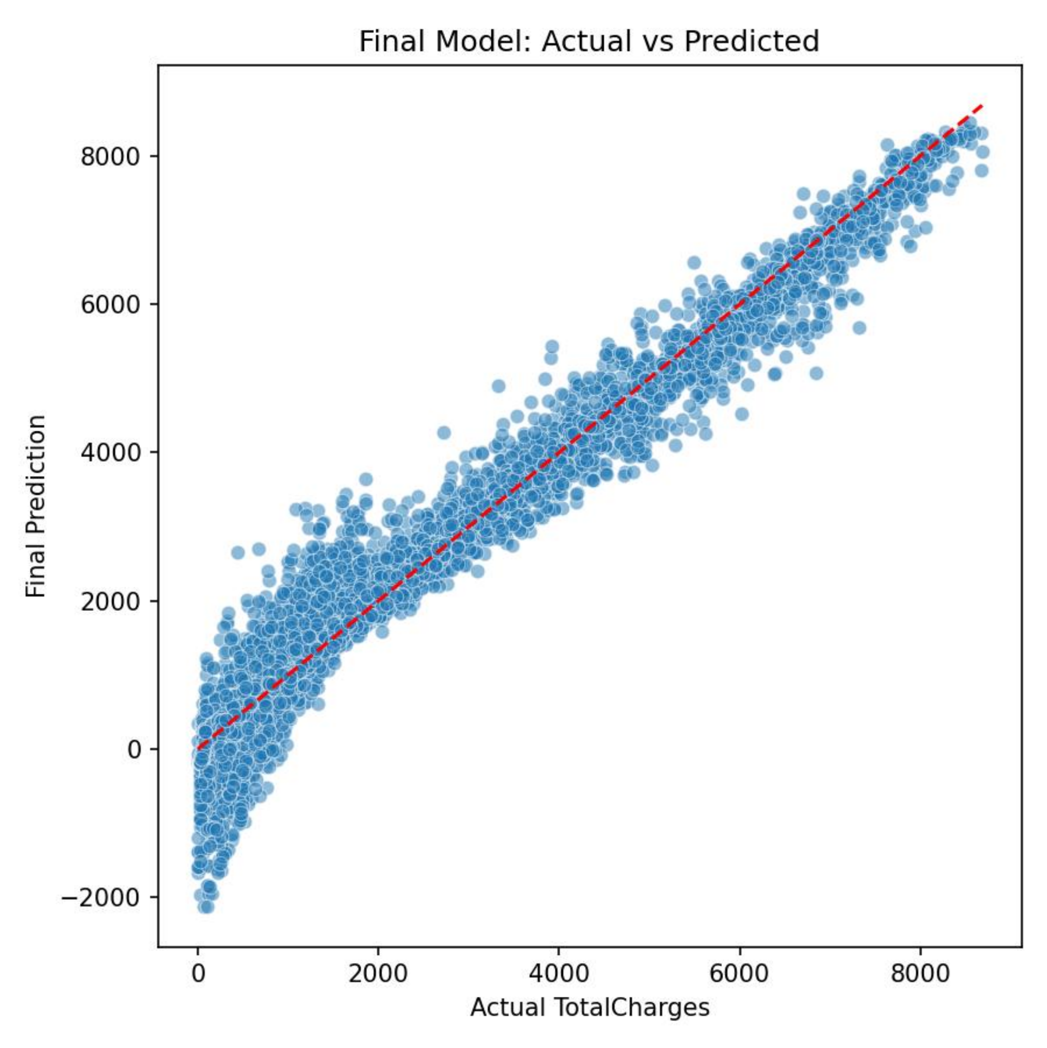
89%
Baseline R^2
Initial linear model performance

55%
Baseline Residual R^2
Initial linear model performance

95.5%
Combined R^2
Enhanced predictive accuracy

18%
Final RMSE





Residual learning improved accuracy compared to simple baseline model

References

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